MATH 202 Your name: _____

Learning target I1, version 1

Compute approximations of definite integrals using numerical methods. Decide when each method gives overestimates vs. underestimates.

Consider the function $f(x) = \dfrac{\sin(x)}{x}$, which does not have an elementary antiderivative. Define the integral function $F(t) = \displaystyle\int_1^t f(x)\,dx$.

(a) What does the Second FTC say about the relationship between F and f?

(b) Use appropriate technology, such as Geogebra or a Riemann sum applet, to approximate $F(2)$ using left, right, midpoint, and trapezoid sums with 10 subintervals:

- $L_{10} =$
- $R_{10} =$
- $M_{10} =$
- $T_{10} =$

Report your answers with at least 6 decimal places of accuracy.

(c) Which of these approximations are overestimates and which are underestimates? How can you tell?

(d) Use your calculations from part (b) to approximate $F(2)$ using Simpson's rule.

MATH 202 Your name: _____

Learning target I1, version 2

Compute approximations of definite integrals using numerical methods. Decide when each method gives overestimates vs. underestimates.

Consider the function $g(x) = \sqrt{1-x^2}$.

(a) Use Geogebra or Desmos or something to draw a graph of this function.

(b) Based on, like, geometry, what's $\int_{-1}^{1} g(x)\,dx$?

(c) So then we can figure out a good approximation to π if we figure out a good approxmation to $\int_{-1}^{1} g(x)\,dx$. Use appropriate technology, such as Geogebra or a Riemann sum applet, to approximate this area using left, right, midpoint, and trapezoid sums with 10 subintervals:

- $L_{10} =$
- $R_{10} =$
- $M_{10} =$
- $T_{10} =$

Report your answers with at least 6 decimal places of accuracy.

(d) Which of these approximations are overestimates and which are underestimates? How can you tell?

(e) Use your calculations from part (b) to approximate $\int_{-1}^{1} g(x)\,dx$ using Simpson's rule.

(f) And lastly but not leastly, give a good approximation of π based on your work.